เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of semester 1 Year: 2015

Subject: 392133 Physics III Section: 15-16

Date: 15 October 2015 Time: 13.00 – 15.00

Name...ID:..Class:................

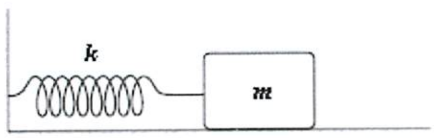
Instructions

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Express your answer in English. Other languages will be discarded.
4. Calculators are **NOT** allowed.
5. Electronic communication device are **NOT** allowed in the examination room.
6. The examination has 7 pages (including this page), and 6 problems in the exam with the total scores of 60 points.
7. Write all your answer on this examination sheet.

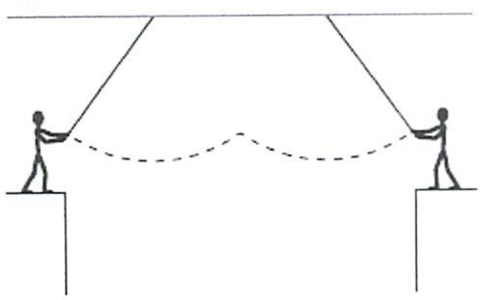
1.1 [5 points] An archer pulls her bow string back 0.40 m by exerting a force that increases uniformly from zero to 240 N. What is the equivalent spring constant of the bow?

1.2 [5 points] An automobile ($m = 1.00 \times 10^3$ kg) is driven into a brick wall in a safety test. The bumper behaves like a spring ($k = 2.50 \times 10^8$ N/m), and is observed to compress a distance of 1.00 cm as the car is brought to rest. What was the initial speed of the automobile?

2.1 [5 points] A mass m = 2.0 kg is attached to a spring having a force constant k = 2 N/m as in the figure. The mass is displaced from its equilibrium position and released. Find its frequency of oscillation (in Hz)



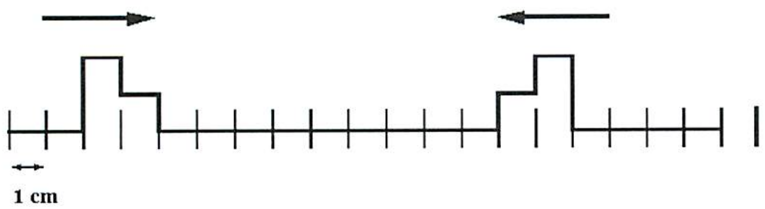
2.2 [5 points] Two circus clowns (each having a mass of 50 kg) swing on two flying trapezes (negligible mass, length 250 m) shown in the figure. At the peak of the swing, one grabs the other, and the two swing back to one platform. Determine the time for the forward and return motion.



3.1 [5 points] A body of mass 5.0 kg is suspended by a spring which stretches 10 cm when the mass is attached. It is then displaced downward an additional 5.0 cm and released. Write down its position as a function of time.

3.2 [5 points] A body oscillates with simple harmonic motion along the x axis. Its displacement varies with time according to the equation $x = 5.0 \cos(\pi t)$. Find The magnitude of the acceleration (in m/s^2) of the body at $t = 1.0$ s.

4.1 [2 points] Two pulses are traveling towards each other at 10 cm/s on a long string at t = 0 s, as shown below. Using the principle of linear superposition, draw the shape of the string's pulses at t = 0.5 s.



4.2 [4 points] A circus performer stretches a tightrope between two towers. He strikes one end of the rope and sends a wave along it toward the other tower. He notes that it takes 1.0 s for the wave to travel the 20 m to the opposite tower. If one meter of the rope has a mass of 0.5 kg, find the tension in the tightrope.

4.3 [4 points] A student wants to establish a standing wave on a wire 1.8 m long clamped at both ends. The wave speed is 540 m/s. What is the minimum frequency she should apply to set up standing waves?

5.1 [2 points] A point source emits sound with a power output of $4\pi \times 100$ watts. What is the intensity (in W/m^2) at a distance of 10.0 m from the source?

5.2 [3 points] By what factor will an intensity change when the corresponding sound level increases by 3 dB? Note: $10^{0.3}=1.99$

5.3 [5 points] A truck moving at 60 m/s passes a police car moving at 40 m/s in the opposite direction. If the frequency of the siren is 500 Hz relative to the police car, what is the change in frequency (in Hz) heard by an observer in the truck as the two vehicles pass each other? (The speed of sound in air is 340 m/s.)

6.1 [5 points] An observer stands 3 m from speaker A and 4 m from speaker B. Both speakers, oscillating in phase, produce 170 Hz waves. The speed of sound in air is 340 m/s. What is the phase difference (in radians) between the waves from A and B at the observer's location, point P?

6.2 [5 points] A vertical tube one meter long is open at the top. It is filled with 75 cm of water. If the velocity of sound is 344 m/s, what will the fundamental resonant frequency be (in Hz)?

Asst. Prof. Dr. Amarin Ratanavis